SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA0504

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 40%

QUESTIONS		
Q. No	**Question**	**Bloom's Level**
1	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember
2	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand
3	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand
4	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand
5	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember

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Code of Conduct:

I ***<u>SANTHOSH E / 192511470</u>*** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
KARNAM HARSHITH

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.

Schema:
A schema is the overall structure or blueprint of a database. It defines how data is organized into tables, attributes, relationships, and constraints. A schema changes very rarely.

Example:
A Student table with attributes:

Student_ID

Name

Department

Age

This table design is called the database schema.

Instance

An instance is the actual data stored in the database at a particular moment. It changes whenever records are inserted, updated, or deleted.

Example:

Student_ID	Name	Department	Age
101	Rahul	CSE	19
102	Priya	ECE	20

This data is an instance of the Student schema.

Difference between Schema and Instance:

Schema	Instance
Blueprint of database	Actual data stored
Changes rarely	Changes frequently
Defines structure	Contains records

Types of Schema:
1. Logical Schema:

Describes the logical structure of the database.

Specifies tables, attributes, relationships, and constraints.

Independent of physical storage.

Example: Student(Student_ID, Name, Department, Age)

2. Physical Schema:

Describes how data is physically stored.

Includes storage location, indexes, file organization, and access methods.

Example: Student records stored in B+ Tree index on SSD.

3. View Schema (External Schema):

Shows only the required portion of the database to users.

Enhances security by hiding unnecessary data.

Example: A faculty member can view only Student_ID, Name, and Department but not marks or fees.

2. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

Entities and Attributes

Customer:

- Customer_ID (Primary Key)
- Name
- Address
- Phone
- Email

Account

- Account_No (Primary Key)
- Account_Type
- Balance

Branch

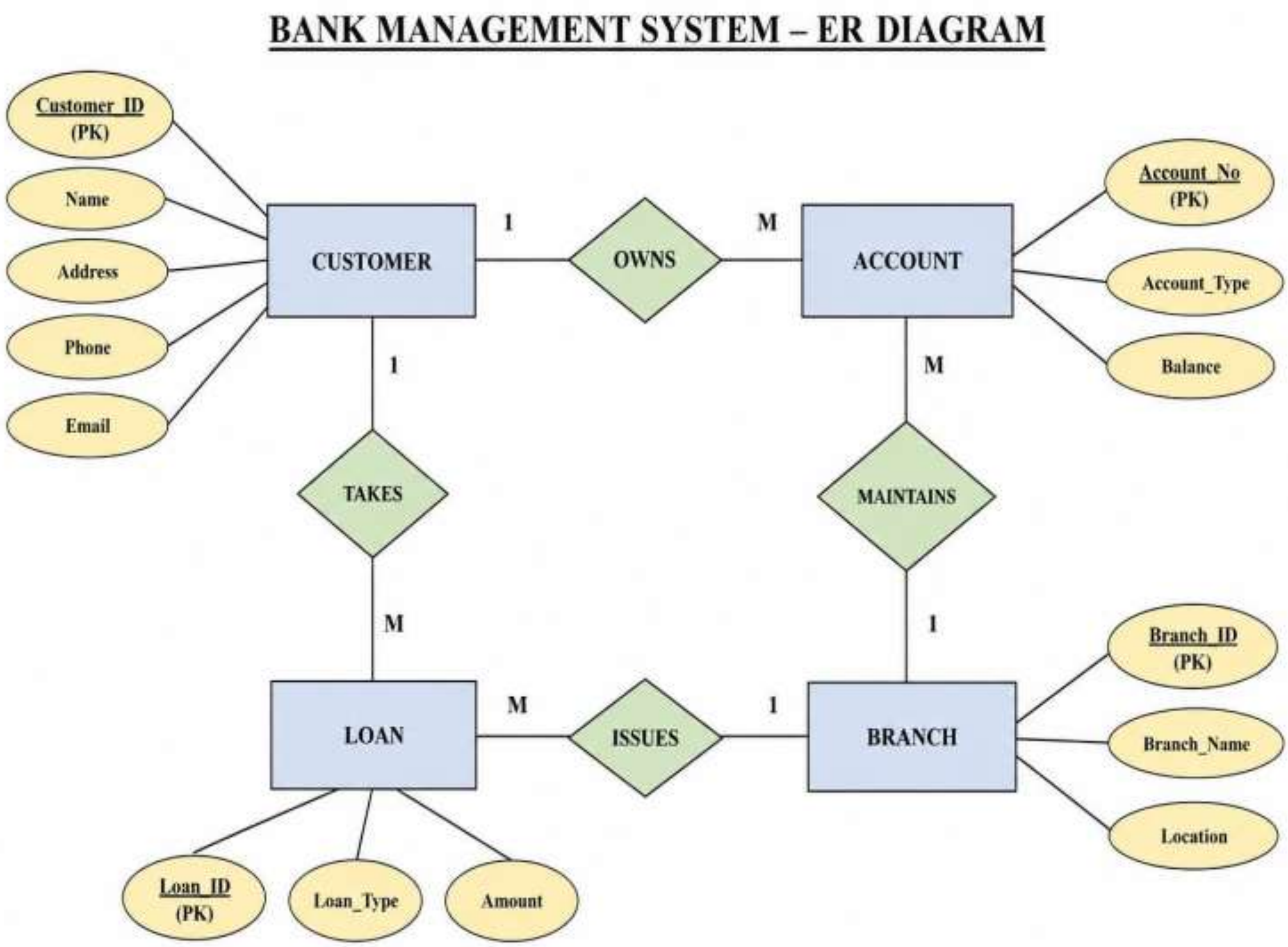
- Branch_ID (Primary Key)
- Branch_Name
- Location

Loan

- Loan_ID (Primary Key)
- Loan_Type
- Amount

Relationships

- Customer owns Account (1:M)
- Branch maintains Account (1:M)
- Customer takes Loan (1:M)
- Branch issues Loan (1:M)



3. Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.

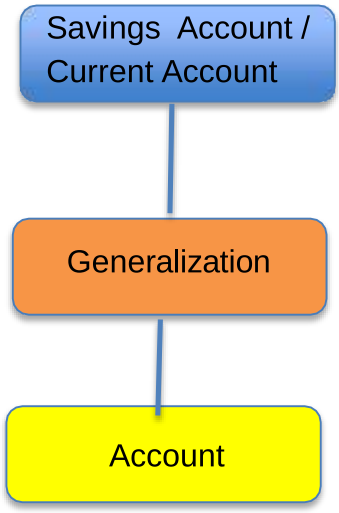
Extended ER (EER) Model

The Extended Entity-Relationship (EER) Model is an enhancement of the ER model. It supports advanced concepts like inheritance, specialization, generalization, and aggregation to represent complex real-world applications.

1. Generalization

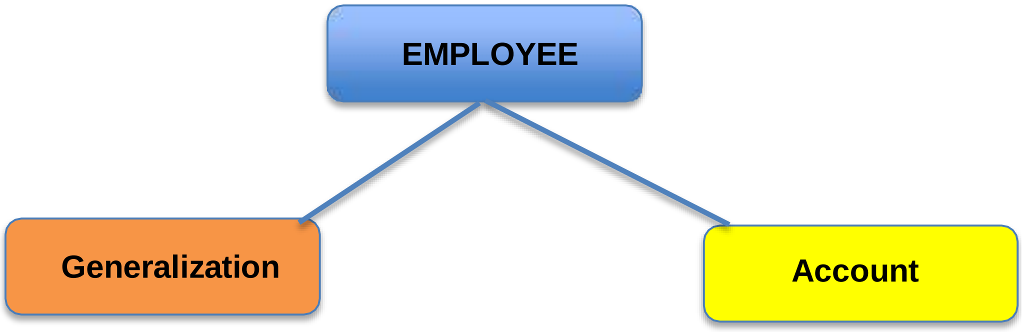
Generalization combines multiple lower-level entity types into a higher-level entity based on common characteristics.

Example:



2. Specialization
Specialization divides a higher-level entity into lower-level entities based on distinguishing characteristics.

Example:



Both Doctor and Nurse inherit Employee attributes.

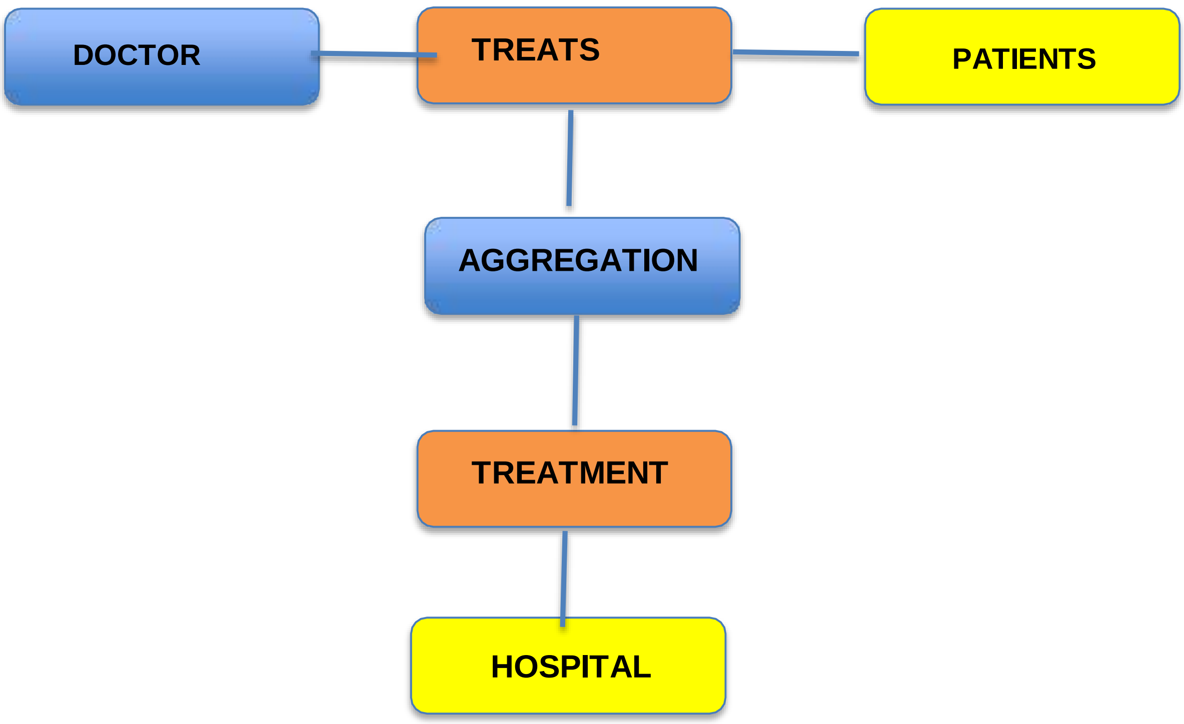
3. Aggregation
Aggregation treats a relationship as a higher-level entity so that it can participate in another relationship.

Example:

A Doctor treats a Patient.

This "Treats" relationship can be aggregated into a new entity called Treatment, which is then related to Hospital.

Doctor ---- Treats ----



4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.

Strong Entities

Patient

- Patient_ID (PK)
- Name
- Age
- Gender

Doctor

- Doctor_ID (PK)
- Name

- Specialization

Hospital

- Hospital_ID (PK)
- Name
- Address.

Weak Entity

Dependent

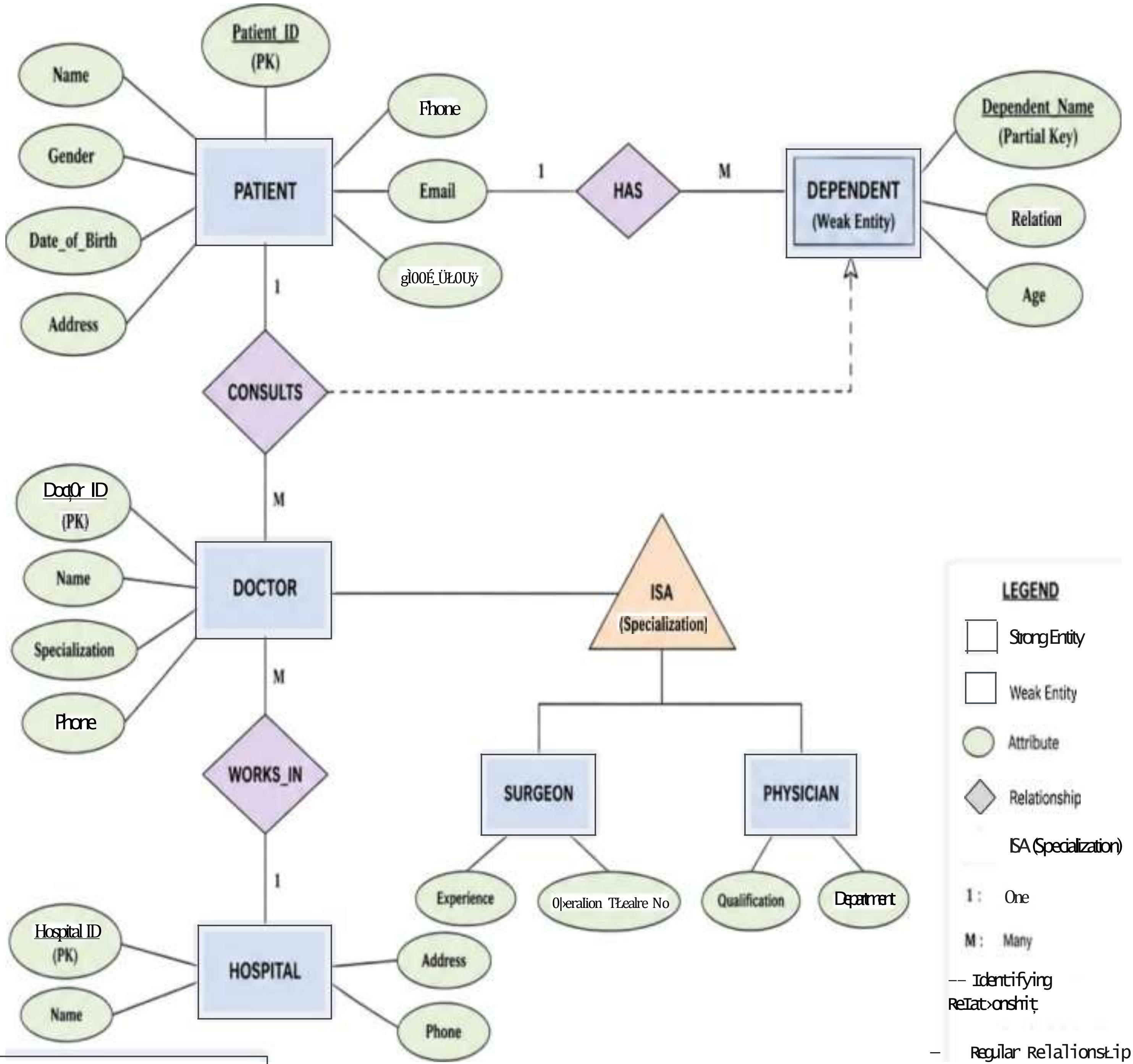
- Dependent_Name (Partial Key)
- Relation
- Age

Dependent cannot exist without Patient.

Relationships

- Patient consults Doctor
- Doctor works in Hospital
- Patient has Dependent (Weak Entity)

E£R DIAGILAM FOR HOSPITAL MANAGEMENT SYSTEM



Notes:

- DEPENDENT is a weak entity and cannot exist without PATIENT.
- DOCTOR is specialized into fLRGE0.1 and PHYfICIAX (ISO hierarchy).

5. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

Database Administrator (DBA)
A Database Administrator (DBA) is responsible for managing, maintaining, securing, and ensuring the efficient operation of databases in an organization.

Responsibilities

1. Install and configure DBMS software.
2. Create and maintain databases.
3. Manage users and assign privileges.
4. Ensure data security and privacy.
5. Perform backup and recovery operations.
6. Monitor database performance and optimize queries.
7. Maintain data integrity and consistency.
8. Handle concurrency control and transaction management.
9. Troubleshoot database issues.
10. Upgrade and patch database systems.

Conclusion

A DBA ensures that databases are secure, reliable, efficient, available, and recoverable, enabling smooth operation of organizational applications.

THANK YOU